How does the electromagnetic train work?

Electromagnets are used in Electromagnetic train. Electromagnets are magnetised only when current flows through them. When the direction of current is changed the poles of the electromagnets are also changed. Like poles of the magnets which are attached at the bottom of the train and rail track repel each other. So, the train is lifted from the track up to a height of 10 cm.

We Know that we can move any magnetic object with the force of attraction or repulsion properties of magnets. This train also moves with the help of the magnets attached on the sides of track and the magnets fitted at the bottom sideway of the train. By controlling the current we can control the magnets and movement of the train.

As there are no moving parts, there is no friction. So, the train can easily attain a speed of 300 km per hour. These trains are capable of running up to 600 km/ hour. They do not make any noise. They require less energy and they are eco-friendly.

Even though, many countries have taken effort to use these trains, such trains are used for public transport only in China, Japan and South Korea. In India the possibilities of introducing these trains are under consideration.

Write the differences between a normal train and an electromagnetic train.

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Points to remember

- Magnetites are natural magnets. They are called magnetic stones.
- Man-made magnets are called artificial magnets.
- Substances which are attracted by magnet are called as magnetic substances.
- Substances which are not attracted by magnet are called non-magnetic substances.
- A freely suspended magnet always comes to rest in north-south direction.
- The end of the magnet that points to the north is called the North Pole. The end that points to the south is called the South Pole.
- A compass is an instrument which is used to find directions.
- Like Poles (N-N, S-S) repel each other and unlike poles (N-S, S-N) attract each other.
- Magnets lose their properties if they are heated or dropped from a height or hit with a hammer.

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